

Time-Domain & Multi-Messenger Astronomy 2026

Day 1 Practical Session: Interactive Alert Broker Triage

Duration: 60 Minutes (20m Demo + 40m Lab) **Session Date:** Day 1 (17:00 UTC+6 / Dhaka Time)

Overview & Learning Objectives

With next-generation optical survey facilities such as the Vera C. Rubin Observatory (LSST) producing millions of real-time alerts per night, automated alert brokers serve as the critical gateway between alert generation and spectroscopic follow-up. This hands-on practical session equips participants with analytical, programmatic, and decision-making skills to triage raw alert streams, evaluate light-curve metrics, eliminate false positives, and trigger follow-up resources.

By the end of this tutorial, participants will be able to:

1. Query the ALERCE broker API to ingest live alert streams from Rubin/LSST.
2. Calculate physical light-curve parameters ($\Delta m / \Delta t$ rise rates and $g - r$ color evolution) to distinguish thermonuclear Supernovae (SNe Ia) from stochastic AGNs.
3. Perform cross-catalog cone searches and query Transient Name Server (TNS) records to eliminate duplicate community targets.
4. Model telescope airmass constraints and generate JSON trigger payloads for automated execution on the Las Cumbres Observatory (LCO) 2-meter robotic telescope network.

Module 1: Necessary Analytics

Key Theoretical & Photometric Equations

1. **Photometric Rise Rate ($\Delta m / \Delta t$):**

$$\frac{\Delta m}{\Delta t} = \frac{m(t_2) - m(t_1)}{t_2 - t_1} \quad [\text{mag/day}] \quad (1)$$

Young Type Ia Supernovae exhibit rapid cooling expanding photospheres resulting in steep early-time rise rates ($\Delta m / \Delta t \leq -0.10$ mag/day in optical g/r bands).

2. **Apparent $g - r$ Color Index:**

$$(g - r)(t) = m_g(t) - m_r(t) \quad (2)$$

As a supernova photosphere expands and cools, its peak wavelength shifts redward, causing the $(g - r)$ color index to monotonically increase over the first 20 days post-explosion.

3. **Plane-Parallel Airmass Relation (X):**

$$X(t) \approx \sec(z(t)) = \frac{1}{\sin(a(t))} \quad (3)$$

where $z(t)$ is the zenith angle and $a(t)$ is the target altitude angle above the horizon. Spectroscopic follow-up requires $X(t) \leq 1.5$ to minimize atmospheric refraction and extinction.

4. Spectroscopic Exposure Scaling Equation:

$$t_{\text{exp}} = t_0 \cdot 10^{0.4(m_r - m_0)} \cdot \left(\frac{\text{SNR}}{\text{SNR}_0} \right)^2 \quad (4)$$

where t_0 is the baseline exposure time for a reference magnitude m_0 achieving a baseline Signal-to-Noise Ratio (SNR_0) on a 2-meter class telescope (e.g., LCO 2m / FLOYDS spectrograph).

Module 2: Graphical Interpretation & Impostor Filtering

Alert streams contain false positives including Active Galactic Nuclei (AGN), variable stars (e.g., Cataclysmic Variables), and image artifacts. Distinguishing genuine supernovae requires evaluating historical variability baselines, light-curve symmetry, and color evolution.

Table 1: Photometric Features for Transient Classification vs. Common Impostors

Transient Class	Light-Curve Morphology	Historical Baseline	Color Evolution ($g - r$)
Supernova Ia (SN Ia)	Rapid rise (15–20d), parabolic peak, monotonic decay	Clean baseline; 0 detections prior to outburst	Blue-to-red reddening due to iron-peak cooling
Active Galactic Nucleus (AGN)	Stochastic, non-periodic flares; long-term baseline noise	Continuous detections over hundreds/thousands of days	Achromatic; color remains nearly constant
Cataclysmic Variable (CV)	Rapid rise (< 2d), asymmetric sharp outbursts, recurrent	Multi-epoch historical outbursts recorded	Very blue during outburst ($(g - r) \sim 0$)

Task 2.1: Conceptual Triage & Rule Set Verification

Refer to the three candidate light curves plotted in workspace and apply the **Three Triage Rules**:

- The Baseline Rule:** Reject any candidate with detections extending > 100 days prior to the current alert window (flags stochastic AGN or long-period variable stars).
- The Outburst Symmetry Rule:** Reject candidates exhibiting multiple historical brightness spikes over 1000 days.
- The Color Cooling Rule:** Select candidates exhibiting clear color separation between g and r filters during peak brightness.

War Room Exercise: Impostor Rejection

Candidate A has an ALeRCE Machine Learning $P(\text{SN Ia}) = 0.82$, but its light curve shows 15 detection epochs spanning back 600 days. Candidate B has $P(\text{SN Ia}) = 0.68$, a flat historical baseline with zero detections prior to 10 days ago, and a steep brightening profile ($\Delta m_g / \Delta t = -0.15$ mag/day).

Question: Which object is the true Supernova Ia candidate, and why did the ML classifier assign a high probability to the false candidate?

Module 3: Computational Notebook & Python Pipeline (30 Mins)

Execute a complete data triage pipeline using a pre-configured environment to ingest fresh alerts, compute light-curve metrics, cross-match against TNS, and send a robotic telescope payload.

Repository Setup & Environment Execution

Clone the official workshop repository and navigate into the Alert Brokers working directory:

```
git clone https://github.com/cassaiub/TDMMA-2026.git
cd TDMMA-2026/Day1/Alert_Brokers
pip install -r requirements.txt
jupyter lab
```

Subtask 1: Alert Ingestion

Open `notebooks/01.Alert_Triage_Master.ipynb`. Implement an automated ALerCE query using `alerce.query_objects()` to filter the alert stream relative to the current Modified Julian Date (`current_mjd`).

Filtering Criteria to Code:

- Temporal Window: Alerts detected within $\text{MJD}_{\text{current}} - 1 \leq \text{MJD} \leq \text{MJD}_{\text{current}}$.
- Machine Learning Probability: $P(\text{SN Ia}) \geq 0.65$ using the `lc_classifier`.
- Photometric Magnitude Cut: $m_g \leq 19.50$ mag.
- Quality Filter: Minimum detection count $N_{\text{det}} \geq 2$ to eliminate single-epoch artifacts.

Subtask 2: Photometric Metric Calculation

Pass the surviving candidate Object IDs (OIDs) into `alerce.query_lightcurve()`. Write a Python loop to:

1. Plot multi-band (g and r) light curves with photometric error bars using `matplotlib`.
2. Programmatically compute the early rise rate $\frac{\Delta m}{\Delta t}$ over the last 5 days.
3. Filter out targets with historical detections older than 100 days.

Subtask 3: Cross-Catalog Association

To prevent re-observing targets already published by the astronomical community:

1. Extract the Right Ascension (RA) and Declination (Dec) coordinates for short-listed candidates.
2. Execute the built-in function `cassa.tns_check(oid, ra, dec)`. This queries the Transient Name Server (TNS) database.
3. Identify the target that returns "UNCLASSIFIED" (rejecting targets labeled as previously spectroscopically confirmed).

Subtask 4: Target Observability

Use `astropy.coordinates` and `astropy.time` to evaluate observability from the LCO Cerro Tololo site.

1. Compute the airmass trajectory $X(t)$ across the upcoming night. Verify $X(t) \leq 1.5$.
2. Calculate the required exposure time t_{exp} using Equation (4).
3. Format your final prioritized candidate selection into a JSON observation trigger payload and execute `submit_lco_trigger(payload)`.

Appendix: Glossary

ALeRCE	Automatic Learning for the Rapid Classification of Events. An astronomical alert broker developed in Chile that ingests wide-field survey streams and applies machine learning classifiers.
Rubin Observatory / LSST	Legacy Survey of Space and Time conducted by the Vera C. Rubin Observatory in Chile. A 8.4-meter wide-field optical survey generating ~ 10 million alerts per night.
ZTF	Zwicky Transient Facility. A wide-field time-domain optical survey operating at Palomar Observatory, acting as a technical precursor to LSST.
TNS	Transient Name Server. The official International Astronomical Union (IAU) mechanism for logging and tracking newly discovered astronomical transients and their spectroscopic classifications.
GCN	General Coordinates Network. A global satellite and ground network system that distributes urgent alerts and follow-up observations of high-energy time-domain transients.
LCO 2m Network	Las Cumbres Observatory 2-Meter Network. A globally distributed, fully automated network of 2-meter robotic telescopes optimized for rapid spectroscopic follow-up.
MJD	Modified Julian Date. A continuous count of days defined as $\text{MJD} = \text{JD} - 2400000.5$, widely used in time-domain astronomy to eliminate timezone ambiguity.
Airmass (X)	A measure of the amount of Earth's atmosphere encountered by optical light from a celestial target, defined relative to the zenith ($X = 1.0$).
AGN	Active Galactic Nuclei. Accreting supermassive black holes at the centers of galaxies that exhibit erratic stochastic optical variability, commonly acting as false positives for SNe Ia.
Type Ia Supernova	A thermonuclear explosion of a carbon-oxygen white dwarf in a binary system, serving as standardizable candles for cosmological distance measurements.